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Collatz Conjecture Tools - Python Execution Manual

1. Introduction

This manual explains how to run the following Python programs (Flask web applications) related to the Collatz Conjecture.

Program Name	Description
`Collatz_Launcher.py`	**(Recommended)** Unified launcher combining all 5 tools below into a single application with a menu
`Counting_by_Length.py`	Counts converged/unconverged Parity Vectors (PV) by length
`Counting_by_Hamming_Weight.py`	Counts converged/unconverged PVs by Hamming weight
`Collatz_Calculator.py`	Displays the Collatz operation process for any natural number
`Collatz_PV_Graph_from_String.py`	Graphical display of an input Parity Vector
`Collatz_PV_Graph_from_Number.py`	Graphical display of the PV generated from an input natural number N

All of these are **Flask** web applications. They are operated through a web browser.

> **Tip:** `Collatz_Launcher.py` combines all five tools into one program with a menu bar, so you only need to run a single file and use one browser tab. The five individual programs remain available separately if you prefer to run them independently.

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2. Prerequisites

2.1 Required Software

Software	Recommended Version	How to Check
Python	3.8 or later	`python --version`
pip	Included with Python	`pip --version`

If Python is not installed, download it from https://www.python.org/downloads/.

> **Note for Windows users:** Be sure to check the “Add Python to PATH” checkbox during installation.

2.2 Installing Flask

Open Command Prompt (Windows) or Terminal (Mac/Linux) and run:

```
pip install flask
```

To check whether Flask is already installed:

```
pip show flask
```

> Once installed, Flask can be used by all five programs above. No need to reinstall for each one.

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3. Setting Up the Programs

3.1 Create a Folder

Create a working folder anywhere on your computer. It is recommended to avoid paths containing non-English characters or spaces.

Windows example:

```
C:\collatz\
```

Mac / Linux example:

```
/home/username/collatz/
```

3.2 Place the Files

Put the downloaded `.py` files into this folder.

```
collatz/
├── Collatz_Launcher.py                 (recommended: all-in-one)
├── Counting_by_Length.py
├── Counting_by_Hamming_Weight.py
├── Collatz_Calculator.py
├── Collatz_PV_Graph_from_String.py
└── Collatz_PV_Graph_from_Number.py
```

> **Note:** If you only plan to use `Collatz_Launcher.py`, the five individual program files are not required — the launcher contains all of their functionality internally. If you want to run multiple individual programs simultaneously, place them in separate folders, or change the port number for each (see "Changing the Port Number" below), since they all default to port 5000.

The `data/` folder will be **automatically created** next to each program when it runs. You do not need to create it manually.

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4. How to Run the Programs

4.1 Basic Steps

1. Open Command Prompt (or Terminal)
2. Navigate to the folder containing the program (using the `cd` command)
3. Run the program with the `python` command
4. Open the URL in your browser

4.2 Recommended: Running the Unified Launcher

The simplest way to use all five tools is to run the launcher, which provides a menu bar for switching between them.

```
cd C:\collatz
python Collatz_Launcher.py
```

If successful, you will see a message similar to:

```
 * Serving Flask app 'Collatz_Launcher'
 * Debug mode: on
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://0.0.0.0:5000
```

Open your browser and go to:

```
http://localhost:5000
```

You will see a top menu bar with the following items. Click any item to switch tools without restarting the program:

- **Home**
- **1. Counting by Length**
- **2. Counting by Hamming Weight**
- **3. Collatz Calculator**
- **4. PV Graph (from PV string)**
- **5. PV Graph (from Number)**

All result files from any tool are saved together in the same `data/` folder next to `Collatz_Launcher.py`.

4.3 Alternative: Running Individual Programs

If you prefer to run a single tool on its own (for example, `Counting_by_Length.py`):

```
cd C:\collatz
python Counting_by_Length.py
```

Open your browser and enter one of the following addresses:

```
http://localhost:5000
```

or

```
http://127.0.0.1:5000
```

If the page displays correctly, the program has started successfully.

4.4 Running the Other Individual Programs

```
cd C:\collatz
python Collatz_Calculator.py
```

```
cd C:\collatz
python Collatz_PV_Graph_from_String.py
```

```
cd C:\collatz
python Collatz_PV_Graph_from_Number.py
```

```
cd C:\collatz
python Counting_by_Hamming_Weight.py
```

In each case, open `http://localhost:5000` in your browser after running the program.

> **Note:** You do not need to run the individual programs if you are using `Collatz_Launcher.py` — it already includes all of their functionality. The individual programs are only needed if you specifically want to run one tool by itself.

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5. Stopping the Program

In the Command Prompt (or Terminal) window where the program is running, press:

```
Ctrl + C
```

(On Mac, this is also `Control + C`.)

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6. Running Multiple Programs at the Same Time

> **Note:** If you use `Collatz_Launcher.py`, this section does not apply — all five tools run inside a single Flask application on one port, so there is no conflict to resolve. This section is only relevant if you run two or more of the **individual** programs at the same time.

By default, Flask uses **port 5000**. If you try to run two or more individual programs at the same time, you will see an error such as:

```
OSError: [Errno 98] Address already in use
```

Solution: Change the Port Number

Find the following line near the end of each program:

```
app.run(debug=True, host='0.0.0.0', port=5000)
```

Change the port number (e.g., 5001, 5002, 5003, ...):

```
app.run(debug=True, host='0.0.0.0', port=5001)
```

Then access it in your browser using that port number:

```
http://localhost:5001
```

Program	Suggested Port (example)
Counting_by_Length.py	5000
Counting_by_Hamming_Weight.py	5001
Collatz_Calculator.py	5002
Collatz_PV_Graph_from_String.py	5003
Collatz_PV_Graph_from_Number.py	5004

7. Overview of Each Program

7.0 Collatz_Launcher.py (Unified Launcher)

- **Input/Output:** Combines all five tools below (7.1–7.5) into a single application
- **Navigation:** Use the top menu bar to switch between tools without restarting the program
- **Storage:** All result files from any tool are saved in the same shared `data/` folder
- **Recommended** as the default way to use this tool suite

7.1 Counting_by_Length.py

- **Input:** Start length (A), end length (B), ratio precision
- **Output:** A table summarizing converged/unconverged PV counts for each length k
- Result files are saved as `.txt` in the `data/` folder

7.2 Counting_by_Hamming_Weight.py

- **Input:** Start Hamming weight (A), end Hamming weight (B)
- **Output:** A table summarizing converged/unconverged PV counts for each Hamming weight d
- Result files are saved as `.txt` in the `data/` folder

7.3 Collatz_Calculator.py

- **Input:** Any natural number (2 or greater)
- **Output:** Each step of the Collatz operation, the Glide value, and the Parity Vector

- If file output is selected, results are saved as `.csv` in the `data/` folder

7.4 Collatz_PV_Graph_from_String.py

- **Input:** A Parity Vector (a string of 0s and 1s)
- **Output:** A graphical matrix of the PV path and a detailed information table
- Detailed result files are saved as `.html` in the `data/` folder

7.5 Collatz_PV_Graph_from_Number.py

- **Input:** Any natural number N (2 or greater)
- **Output:** Automatically computes the PV from N up to its Glide point and displays it graphically
- Uses the same graphical style as `Collatz_PV_Graph_from_String.py`

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8. Troubleshooting

Symptom	Cause and Solution
`python` command not found	Python is not installed, or PATH is not set. Reinstall and check "Add to PATH"
`ModuleNotFoundError: No module named 'flask'`	Flask is not installed. Run `pip install flask`
`Address already in use`	Another program is already using the same port. Change the port number (see Section 6), or stop the running program with `Ctrl+C`
Browser shows nothing	Confirm the URL is correctly entered as `http://localhost:5000` (or with the correct port number). Check for error messages in the Command Prompt
Calculation takes a very long time	The input range may be too large. Check the input field's upper limit (often 10000)
Garbled characters	Confirm your browser's character encoding is set to UTF-8 (usually automatic)

9. Frequently Asked Questions (FAQ)

Q. How do I run the program again after closing it? A. Just repeat the same steps (navigate to the folder and run `python filename.py`).

Q. Where are the calculation result files saved? A. They are saved in the `data/` folder located next to each program.

Q. Do I need an internet connection? A. Only for the initial Flask installation. After that, the programs run entirely offline (locally).

Q. Will this work the same way on other computers? A. Yes. As long as Python and Flask are installed, it works the same way on Windows, Mac, or Linux.

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10. Summary: Quick Command Reference

```
# 1. One-time only: install Flask
pip install flask

# 2. Navigate to the folder
cd C:\collatz

# 3a. Recommended: run the unified launcher (all 5 tools in one menu)
python Collatz_Launcher.py

# 3b. Alternative: run an individual program
python Counting_by_Length.py
python Counting_by_Hamming_Weight.py
python Collatz_Calculator.py
python Collatz_PV_Graph_from_String.py
python Collatz_PV_Graph_from_Number.py

# 4. Open in your browser
http://localhost:5000

# 5. To stop
Ctrl + C
```

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